(c) In another experiment, the solubility of solid organic acid, **W**, in different solvents, water, ethanol and acetone is investigated.

Plan an investigation to find the effect of the nature of solvent on the solubility of organic acid, **W**.

You may assume the apparatus normally found in the school laboratory is available for the investigation.

Approach 1: Add excess solute, find mass not dissolved	• **Apparatus**: electronic mass balance + measuring cylinder/pipette/burette [1] • **Constant variables** (at least 2): volume of solvent, mass of solute used, temperature of solvent [1] • **Procedure 1:** Add a known mass (in excess) of solid organic acid to a known volume of water in a suitable container (beaker/test-tube/boiling tube/cup) + Stir/mix/shake until no more solid appears to dissolve [1] • **Procedure 2**: Filter the mixture to obtain the residue + Dry+ Measure mass of residue [1] • **Independent variable**: Repeat with other solvents, keeping the mass of solid and volume of solvent constant. [1]
Approach 2: Add solute gradually, stop when no more dissolves	• **Apparatus**: electronic mass balance + measuring cylinder/pipette/burette [1] • **Constant variables** (at least 2): volume of solvent, temperature of solvent [1] • **Procedure 1:** Add a known and fixed volume of water into a suitable container (beaker/test-tube/boiling tube/cup). Add solid W in small, measured increments while stirring until all solid dissolves. [1] • **Procedure 2:** Continue adding the solid until no more dissolves. Calculate the total mass of the solid that has dissolved. [1] • **Independent variable**: Repeat with other solvents, keeping the volume of solvent and temperature of solvent constant. [1]
Approach 3: Add excess solute, find mass that dissolves	• **Apparatus**: electronic mass balance + measuring cylinder/pipette/burette [1] • **Constant variables** (at least 2): volume of solvent, mass of solute, temperature of solvent [1] • **Procedure 1:** Add a known mass (in excess) of solid organic acid to a known volume of water in a suitable container (beaker/test-tube/boiling tube/cup) + Stir/mix/shake until no more solid appears to dissolve [1] • **Procedure 2:** Filter + heat filtrate to evaporate **all** water and measure the mass of dissolved solute. [1] • **Independent variable**: Repeat with other solvents, keeping the volume of solvent, mass of solute used and temperature of solvent constant. [1]
Approach 4: Fixed mass solute add solvent until all dissolves	• **Apparatus**: electronic mass balance + measuring cylinder/burette (R: pipette) [1] • **Constant variables** (at least 2): volume of solvent, temperature of solvent [1] • **Procedure 1**: Add known volume of water <u>gradually</u> to known mass of W in a container (beaker/test-tube/boiling tube/cup) + Stir/mix/shake [1] • **Procedure 2:** If not all dissolved, add more water. Calculate the total volume of water added. [1] • **Independent variable**: Repeat with other solvents, keeping the volume of solvent and temperature of solvent constant. [1]